#1. $\dot{x} = Ax + Bu + f(t)$. $x(0) = x_0$.

$J = \frac{1}{2} x^T Q x \big|_{t=t_f} + \frac{1}{2}\int_0^{t_f} (x^T Q x + u^T R u)\,dt$.

Let $u(t) = -R^{-1}B^T(kx + r)$ and $\dot{r}(t) = (kBR^{-1}B^T - A^T)r - k f(t)$.

Define $H = \frac{1}{2}x^T Q x + \frac{1}{2}u^T R u + \lambda^T(Ax + Bu + f(t))$.

$\frac{\partial H}{\partial u} = 0 \Rightarrow \frac{\partial H}{\partial u} = Ru + B^T\lambda = 0$.

$\Rightarrow u = -R^{-1}B^T\lambda$.

$\dot{\lambda} = -\left(\frac{\partial H}{\partial x}\right)^T = -(Q^T x + A x)$

$= -Q^T x - A^T\lambda$

$\lambda(t_f) = Q x(t_f)$.

Define $\lambda = kx + r$ such that $k(t_f) = Q$ and $r(t_f) = 0$.

$\dot{\lambda} = \dot{k}x + k\dot{x} + \dot{r}$

$= \dot{k}x + k(Ax + Bu + f(t)) + (kBR^{-1}B^T - A^T)r - k f(t)$

$= -Q^T x - Ax$

$\Rightarrow \dot{k}x + kAx + kB(-R^{-1}B^T\lambda) + \cancel{kf(t)} + kBR^{-1}B^T r - A^T r \cancel{- kf(t)}$

$= -Q^T x - A(kx + r)$

$\Rightarrow \dot{k}x + kAx - kBR^{-1}B^T kx - \cancel{kBR^{-1}B^T r} + \cancel{kBR^{-1}B^T r} - \cancel{A^T r}$

$= -Q^T x - A^T kx - \cancel{A^T r}$.

$\Rightarrow (\dot{k} + kA - kBR^{-1}B^T k + Q^T + A^T k)x = 0$.

$\therefore \dot{k} + kA - kBR^{-1}B^T k + Q^T + A^T k = 0$

which is Riccati equation.

#2. $\dot{x}_1 = x_2$ & $\dot{x}_2 = -ax_2 - bx_1 + u$.

$J = \frac{1}{2}\int_0^{t_f} [t(c_1 x_1^2 + c_2 x_2^2)]\,dt$.

Define $H = \frac{1}{2}t(c_1 x_1^2 + c_2 x_2^2) + \lambda_1 x_2 + \lambda_2(-ax_2 - bx_1 + u)$.

$\frac{\partial H}{\partial u} = \lambda_2 = 0$.

$\dot{\lambda} = -\left(\frac{\partial H}{\partial x}\right)^T \Rightarrow \begin{cases} \dot{\lambda}_1 = c_1 t x_1 - b\lambda_2 \\ \dot{\lambda}_2 = c_2 t x_2 + \lambda_1 - a\lambda_2. \end{cases}$

$\lambda(t_f) = \left[\frac{\partial\phi}{\partial x_f} + v^T\frac{\partial\psi}{\partial x_f}\right]_{t=t_f} = 0$.

Since there is no equation for $u(t)$, then the optimal law is singular for all time t_f.

$\dot{\lambda}_2 = c_2 t x_2 + \lambda_1 - a\lambda_2 = c_2 t x_2 + \lambda_1$.

$\ddot{\lambda}_2 = c_2 x_2 + c_2 t\dot{x}_2 + \dot{\lambda}_1 = c_2 x_2 + c_2 t(-ax_2 - bx_1 + u) + c_1 t x_1 - b\lambda_2$.

$= c_2 x_2 + c_2 t(-ax_2$

$= c_2 x_2 - a c_2 t x_2 - b c_2 t x_1 + c_2 t u + c_1 t x_1$
$= 0$

$\Rightarrow c_2 t u = (1 + at) c_2 x_2 + (b c_2 - c_1) t x_1$

$\therefore u(t) = \dfrac{b c_2 - c_1}{c_2} x_1 + \dfrac{-1 + at}{t} x_2$

$\Rightarrow \dot{x}_1 = x_2.$
$\dot{x}_2 = -a x_2 - b x_1 + \dfrac{b c_2 - c_1}{c_2} x_1 + \dfrac{1 - at}{t} x_2.$

$= x_1 \left(\cancel{b} - \dfrac{c_1}{c_2} - \cancel{b}\right) + \left(-\cancel{a} - \dfrac{1}{t} + \cancel{a}\right) x_2.$

$= - \dfrac{c_1}{c_2} x_1 - \dfrac{1}{t} x_2.$

$\Rightarrow \ddot{x}_1 = \dot{x}_2 = - \dfrac{c_1}{c_2} x_1 - \dfrac{1}{t} x_2 = - \dfrac{c_1}{c_2} x_1 - \dfrac{1}{t} \dot{x}_1$

$\Rightarrow \ddot{x}_1 + t^{-1} \dot{x}_1 + (c_1 / c_2) x_1 = 0.$

$\Rightarrow t \ddot{x}_1 + \dot{x}_1 + t (c_1/c_2) x_1 = 0$

$\Rightarrow t^2 \ddot{x}_1 + t \dot{x}_1 + t^2 (c_1/c_2) x_1 = 0.$

$\Rightarrow x_1(t) = c_1 J_0\left(\sqrt{\dfrac{c_1}{c_2}}\, t\right) + c_2 Y_0\left(\sqrt{\dfrac{c_1}{c_2}}\, t\right)$ (if $c_1 c_2 > 0$)

where $\begin{cases} J_0 \text{ is the Bessel function of the first kind and} \\ Y_0 \text{ is the Bessel function of the Sencond kind.} \end{cases}$

Since $x_2 = \dot{x}_1$, then we can solve for u.

#3. $\dot{x}_1 = x_2$. $\dot{x}_2 = u_1$, $\dot{x}_3 = x_4$. $\dot{x}_4 = u_2$.

$x(0) = x_0$.

$S(x) = x_1^2 + x_3^2 - 1 = 0.$

$J = \int_0^{t_f} (u_1^2 + u_2^2)\, dt.$

$\dot{S}(x) = 2 x_1 \dot{x}_1 + 2 x_3 \dot{x}_3$

$= 2 x_1 x_2 + 2 x_3 x_4 = 0$

$\ddot{S}(x) = 2 \dot{x}_1 x_2 + 2 x_1 \dot{x}_2 + 2 \dot{x}_3 x_4 + 2 x_3 \dot{x}_4$

$= 2 x_2^2 + 2 x_1 u_1 + 2 x_4^2 + 2 x_3 u_2 = 0$

Define Hamiltonian $H = u_1^2 + u_2^2 + \lambda_1 x_2 + \lambda_2 u_1 + \lambda_3 x_4 + \lambda_4 u_2$
$+ \mu (x_2^2 + x_1 u_1 + x_4^2 + x_3 u_2)$

$\Rightarrow J_a = S(x(t_f)) + \int_0^{t_f} (H - \lambda^T \dot{x})\, dt.$

$\dfrac{\partial H}{\partial u} = 0 \Rightarrow \begin{cases} 2 u_1 + \lambda_2 + \mu x_1 = 0. \\ 2 u_2 + \lambda_4 + \mu x_3 = 0. \end{cases}$

$\Rightarrow \begin{cases} u_1 = -(\lambda_2 + \mu x_1)/2. \\ u_2 = -(\lambda_4 + \mu x_3)/2. \end{cases}$

$0.5\ddot{S}(x) = x_2^2 + x_1(\lambda_2 + \mu x_1)/2 + x_4^2 + x_3(\lambda_4 + \mu x_3)/2 = 0.$

$\Rightarrow 2x_2^2 + 2x_4^2 = \lambda_2 + \lambda_4 + \mu(x_1 + x_3).$

$\Rightarrow \mu = \dfrac{2x_2^2 + 2x_4^2 - x_1\lambda_2 - x_3\lambda_4}{x_1 + x_3} = 2x_2^2 + 2x_4^2 - x_1\lambda_2 - \lambda_4 x_3$

$\lambda(t_f) = \nu^T \dfrac{\partial S}{\partial x_f} \Rightarrow \begin{cases} \lambda_1(t_f) = 2x_1(t_f)\nu_1 \\ \lambda_2(t_f) = 0 \end{cases} \begin{cases} \lambda_3(t_f) = 2x_3(t_f)\nu_3 \\ \lambda_4(t_f) = 0. \end{cases}$

$\dot{\lambda} = -\dfrac{\partial H}{\partial x} \Rightarrow \begin{cases} \dot{\lambda}_1(t) = -\mu u_1 \\ \dot{\lambda}_2(t) = -\lambda_1 \mp 2\mu x_2. \\ \dot{\lambda}_3(t) = -\mu u_2 \\ \dot{\lambda}_4(t) = -\lambda_3 \mp 2\mu x_4 \end{cases}$

Equations.

$x_2^2 + x_1 u_1 + x_4^2 + x_3 u_2 = 0.$

$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = u_1 \\ \dot{x}_3 = x_4 \\ \dot{x}_4 = u_2. \end{cases}$

$\dfrac{\partial H}{\partial u} = 0 \Rightarrow \begin{cases} u_1 = -(\lambda_2 + \mu x_1)/2. \\ u_2 = -(\lambda_4 + \mu x_3)/2. \end{cases}$

$\mu = 2x_2^2 + 2x_4^2 - x_1\lambda_2 - \lambda_4 x_3.$

$\dot{\lambda} = -\dfrac{\partial H}{\partial x} \Rightarrow \begin{cases} \dot{\lambda}_1(t) = -\mu u_1 \\ \dot{\lambda}_2(t) = -\lambda_1 - 2\mu x_2. \\ \dot{\lambda}_3(t) = -\mu u_2. \\ \dot{\lambda}_4(t) = -\lambda_3 - 2\mu x_4. \end{cases}$

Conditions.

$X(0) = X_0.$

$\lambda(t_f) = \nu^T \dfrac{\partial S}{\partial x_f} \Rightarrow \begin{cases} \lambda_1(t_f) = 2x_1(t_f)\nu_1 \\ \lambda_2(t_f) = 0 \\ \lambda_3(t_f) = 2x_3(t_f)\nu_3 \\ \lambda_4(t_f) = 0. \end{cases}$

$x_1(t_f)^2 + x_3(t_f)^2 = 1.$

#4. A direct problem

1) Guess $u(t)$.
2) Forward integrating $\dot{x} = f(x, u, t)$. with initial condition $x(0)$.
3) Backward integrating $\dot{\lambda} = -\frac{\partial H}{\partial x}$ with terminal condition $\lambda(t_f) = \frac{\partial h}{\partial x_f} + \nu^T \frac{\partial \psi}{\partial x_f}$
4) Compute $\delta u(t) = -\alpha \frac{\partial H}{\partial u}$
5) If $|\delta u(t)| < \eta$, then stop computing. If not,
   $u(t) = u^{old}(t) + \delta u(t)$
6) Go to 2).

An indirect problem.

1) Guess $\lambda(t_0)$.
2) Using $\frac{\partial H}{\partial u} = 0$, make u in terms of x and λ. and $\dot{\lambda} = g(x, \lambda, t)$.
3) Forward integrating $\dot{x} = f(x, u(x,\lambda), t)$

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4) Compute $\delta\Lambda(t_0)$

5) If $\delta\Lambda(t_0) < \varepsilon$, then stop iteration

If not, $\Lambda(t_0) = \Lambda^{old}(t_0) + \delta\Lambda(t_0)$.

6) Go to 2).

AST 6037-01 Trajectory Optimization

Final Examination

December 11, 2012 10:00-12:00 (120 minute)

Close Books/Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (20) Consider minimizing the following functional

$$J(x,\dot{x}) = \int_{t_0}^{t_f} L(x,\dot{x},t)dt$$

where $x(t_0)$ and $x(t_f)$ are free.

I. (10) Use the fundamental theorem of the Calculus of Variations to derive the necessary conditions for an extremum.

II. (10) Determine an extremum for $L(x,\dot{x},t) = \frac{1}{2}\dot{x}^2 + x\dot{x} + \dot{x} + x$, $t_0 = 0$ and $t_f = 2$

2. (20) Consider minimizing the performance index

$$J = h\left(x(t_f),t_f\right) + \int_{t_0}^{t_f} g\left(x(t),u(t),t\right)dt$$

subject to the dynamical equations of motion

$$\dot{x}(t) = f(x,u,t)$$

while satisfying the boundary conditions

$$x(t_0) = x_0, \quad m(x(t_f),t_f) = 0$$

where $x \in R^n$, $u \in R^m$, $m \in R^k$, $m \le n$, $k \le n$ and t_f is free. Use the variational principle to derive the first order necessary conditions for optimality.

3. (30) Consider minimizing the performance index

$$J = \int_0^2 (u-1)x dt$$

subject to the dynamics

$$\dot{x} = xu, \; x(0) = x_0, \; x > 0$$

and control bounds

$$0 \le u \le 1$$

I. (10) Use the Pontryagin Principle to derive the first order necessary conditions for optimality.

II. (10) Show that there does not exist any singular interval.

III. (10) Develop a candidate optimal trajectory explicitly for state, costate and control.

4. (10) Consider minimizing the performance index

$$J = \int_{t_0}^{t_f} |u(t)|\, dt$$

subject to the dynamics

$$\dot{x}(t) = u(t), \quad x(t_0) = x_0, \quad x(t_f) = 0, \quad t_f \text{ is free}$$

and control bounds

$$|u(t)| \le 1.0$$

The control objective is for the system to transfer from an arbitrary initial state x_0 to the origin, while minimizing fuel consumption. Use the Pontryagin principle to determine the control logic.

5. (20) Given a function $f : R^n \to R$, suppose that its gradient $G(x_k) \ne 0$ and Hessian $H(x_k) > 0$ at x_k. Show that the direction

$$d_k = -H^{-1}(x_k)G^T(x_k)$$

is a descent direction. That is, show that if

$$x_{k+1} = x_k + \alpha_k d_k$$

then

$$f(x_{k+1}) < f(x_k)$$

for sufficiently small $\alpha_k > 0$.

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#1. $\begin{cases}\dot{x}_1 = x_2 \\ \dot{x}_2 = -x_2 + u\end{cases}$

$\begin{cases} X(0) = X_0 & : \text{Given} \\ t_f & : \text{specified} \\ X_f & : \text{free} \end{cases}$

$|u(t)| \le 1$ for all time t.

$$J = \frac{1}{2}\int_{t_0}^{t_f} (x_1^2 + u^2)\,dt.$$

Define Hamiltonian $H(X, U, \Lambda, t) = \frac{1}{2}x_1^2 + \frac{1}{2}u^2 + \Lambda_1 x_2 + \Lambda_2(-x_2 + u)$

$\frac{\partial H}{\partial u} = 0 \Rightarrow \frac{\partial H}{\partial u} = u + \Lambda_2 = 0. \quad \therefore u(t) = -\Lambda_2(t).$

$\dot{\Lambda} = -\frac{\partial H}{\partial x} \Rightarrow \begin{cases}\dot{\Lambda}_1(t) = x_1(t) \\ \dot{\Lambda}_2(t) = \Lambda_1(t) - \Lambda_2(t)\end{cases}$ If $|u(t)| \le 1$

If the control constraint is acting, $(|u(t)| = 1)$.

From $H = 0.5x_1^2 + 0.5u^2 + \Lambda_1 x_2 - \Lambda_2 x_2 + \Lambda_2 u$,

If $\Lambda_2 > 0$, $u = -1$. and If $\Lambda_2 < 0$, $u = 1$.

$$\Rightarrow u(t) = \begin{cases} 1 & \text{for } \Lambda_2(t) \le -1 \\ -1 & \text{for } \Lambda_2(t) \ge 1 \\ -\Lambda_2(t) & \text{for } |\Lambda_2(t)| < 1 \end{cases}$$

$$\Lambda(t_f) = \left(\frac{\partial \phi}{\partial x_f} + \nu^T \frac{\partial m}{\partial x_f}\right)_{t=t_f} = 0. \Rightarrow \begin{cases}\Lambda_1(t_f) = 0 \\ \Lambda_2(t_f) = 0.\end{cases}$$

Equations

$$\begin{cases} \dot{x}_1 = x_2. \\ \dot{x}_2 = -x_2 + u. \\ u = \begin{cases} 1 & \forall\ \Lambda_2 \le -1 \\ -1 & \forall\ \Lambda_2 \ge 1 \\ -\Lambda_2 & \forall\ |\Lambda_2| < 1 \end{cases} \\ \dot{\Lambda}_1 = x_1 \\ \dot{\Lambda}_2 = \Lambda_1 - \Lambda_2. \end{cases}$$

Conditions

$$\begin{cases} X(0) = X_0. \\ \Lambda(t_f) = 0. \end{cases}$$

#2. Let $x_1 = x$. $x_2 = \dot{x}$, $x_3 = m$, $u = \dot{m}$, then

$\hookrightarrow -M \le u \le 0.$

2-1) $\begin{cases}\dot{x}_1 = x_2. \\ \dot{x}_2 = -g - ku/x_3. \\ \dot{x}_3 = u.\end{cases}$

$$J = \int_0^{t_f} 1\,dt.$$

$$H = 1 + \Lambda_1 x_2 + \Lambda_2(-g - ku/x_3) + \Lambda_3 u$$
$$= 1 + \Lambda_1 x_2 - g\Lambda_2 + (\Lambda_3 - \Lambda_2 k/x_3)u$$

If $-M < u < 0$,

$\frac{\partial H}{\partial u} = 0. \Rightarrow \Lambda_3 - \frac{\Lambda_2 k}{x_3} = 0.$: singular.

If the control constraint is active, H is minimum by Pontryagin's principle
$H = 1 + \lambda_1 x_2 - \lambda_2 g + (\lambda_3 - \lambda_2 k / x_3) u$.
If $\lambda_3 - \lambda_2 k / x_3 > 0$, $u = -M$.
If $\lambda_3 - \lambda_2 k / x_3 < 0$, $u = 0$.

$$\Rightarrow u(t) = \begin{cases} -M & \text{for } \lambda_3 - \lambda_2 k/x_3 > 0 \\ 0 & \text{for } \lambda_3 - \lambda_2 k/x_3 < 0 \\ \text{undefined} & \text{for } \lambda_3 - \lambda_2 k/x_3 = 0 \end{cases}$$

$$\dot{\lambda} = -\left(\frac{\partial H}{\partial x}\right)^T \Rightarrow \begin{cases} \dot{\lambda}_1(t) = 0 & \Rightarrow \lambda_1(t) = \bar{\lambda}_1 \ : \text{const} \\ \dot{\lambda}_2(t) = \lambda_1 & \Rightarrow \lambda_2(t) = \bar{\lambda}_1 t + \bar{\lambda}_2 \\ \dot{\lambda}_3(t) = (\lambda_2 k / x_3^2) u \cdots (A) \end{cases}$$

Since t_f is free,

$$H(t_f) = -\left[\frac{\partial \phi}{\partial t_f} + \nu^T \frac{\partial \vec{m}}{\partial t_f}\right]_{t=t_f} = 0.$$

$$\Rightarrow 1 + \lambda_1(t_f) x_2(t_f) - \lambda_2(t_f) g + (\lambda_3(t_f) - \lambda_2(t_f) k / x_3(t_f)) u(t_f) = 0$$

2-2) Let assume that $\lambda_3 - \lambda_2 k / x_3 = 0$ for a finite time.

$$\Rightarrow H = 1 + \lambda_1 x_2 - \lambda_2 g \ \& \ \frac{d}{dt}(\lambda_3 - \lambda_2 k / x_3) = 0.$$

From (A), $\dot{\lambda}_3(t) - \lambda_2 k / x_3^2 u = 0$.

$$\Rightarrow \frac{d}{dt}(\lambda_3 - \lambda_2 k / x_3) = \dot{\lambda}_3 - \dot{\lambda}_2 k / x_3 + (\lambda_2 k / x_3^2) u = 0.$$

$$\Rightarrow 2\dot{\lambda}_3 - \dot{\lambda}_2 k / x_3 = 0.$$

$$\Rightarrow 2\dot{\lambda}_3 x_3 = \bar{\lambda}_1 k$$

$$\Rightarrow \dot{\lambda}_3 = \frac{\bar{\lambda}_1 \cancel{k}}{2 \cancel{x_3}} = \frac{\lambda_2 \cancel{k}}{x_3^{\cancel{2}}} u$$

$$\Rightarrow \bar{\lambda}_1 x_3 = 2\lambda_2 u \cdots (A)$$

From $H = 1 + \lambda_1 x_2 - \lambda_2 g = 0$.
If $\lambda_1 \neq 0$, $x_2 = (\lambda_2 g - 1)/\lambda_1 = gt + (\lambda_{20}/\bar{\lambda}_1) g - 1/\bar{\lambda}_1$

$$\dot{x}_2 = g = -g - ku/x_3$$

$$\Rightarrow 2g = -ku/x_3$$

$$\Rightarrow 2g x_3 = -ku \cdots (B)$$

$$\Rightarrow -\frac{2g}{k} = \frac{\bar{\lambda}_1}{2\lambda_2} = \frac{\bar{\lambda}_1}{2\bar{\lambda}_1 t + \lambda_{20}} \cdots (C)$$

(C) is satisfied for any $t \geq 0$, then 는 contradiction.
Thus $\bar{\lambda}_1 = 0$, thus (A) becomes $0 = 2\lambda_2 u$.
$\Rightarrow \bar{\lambda}_2 = 0$ OR $u = 0$.
From $H = 0$, $H = 1 - \bar{\lambda}_2 g = 0 \Rightarrow \lambda_2 = g^{-1}$.
Thus $u = 0$. $x_3 = x_{30}$. $\lambda_3 = k/(x_{30} g)$.

$$\Rightarrow \dot{x}_2 = -g, \quad \dot{x}_1 = x_2 \Rightarrow \ddot{x} = -g$$

$$\Rightarrow x = -\frac{1}{2} g t^2 + x_{20} t + x_{10}.$$

It is 자유 낙하 운동
⇒ 자유낙하 운동일때 u가 singular 이다.

#3. $\dot{x} = u$. $x(0) = x_0$ $x(1) = 0$.

$$J = \int_0^1 g(x, u)\,dt.$$

$$x(t_1^+) = x(t_1^-) + 1.$$

$$J_a = \nu(x(t_1^+) - x(t_1^-) - 1) + \int_0^{t_1^-} g(x, u)\,dt + \int_{t_1^+}^1 g(x, u)\,dt.$$

Define $\begin{cases} \text{Hamiltonian } H = g + \lambda^T u. \\ \psi = x(t_1^+) - x(t_1^-) - 1 = 0 \end{cases}$

$$\Rightarrow J_a = \nu\psi + \int_0^{t_1^-} (H - \lambda^T \dot{x})\,dt - \int_1^{t_1^+} (H - \lambda^T \dot{x})\,dt.$$

$$\Rightarrow \delta_a = \nu \frac{\partial \psi}{\partial x(t_1^+)} \delta x(t_1^+) + \nu \frac{\partial \psi}{\partial x(t_1^-)} \delta x(t_1^-) + [H - \cancel{\lambda^T \dot{x}}]\,\delta t_1^-$$

$$- [H - \cancel{\lambda^T \dot{x}}]\,\delta t_1^+ - \cancel{[\lambda^T\ ]\,\delta x(t_1^-)} + \cancel{[\lambda^T\ ]\,\delta x(t_1^+)}$$

$$+ \int_0^{t_1^-} \left\{ \frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial x} + \dot{\lambda}\right)\delta x + \left(\frac{\partial H}{\partial \lambda} - \dot{x}\right)\delta \lambda \right\} dt$$

$$- \int_1^{t_1^+} \left(\frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial x} + \dot{\lambda}\right)\delta x + \left(\frac{\partial H}{\partial \lambda} - \dot{x}\right)\delta \lambda \right\} dt.$$

$$= \left[\nu \frac{\partial \psi}{\partial x(t_1^+)} + \lambda(t_1^+)\right]\delta x_1^+ + \left[\nu \frac{\partial \psi}{\partial x(t_1^-)} - \lambda(t_1^-)\right]\delta x_1^-$$

$$+ \left[H(t_1^-) - H(t_1^+)\right]\delta t_1$$

$$+ \int_0^{t_1^-} \left\{ \left(\frac{\partial H}{\partial u}\right)\delta u + \left(\frac{\partial H}{\partial x} + \dot{\lambda}\right)\delta x + \left(\frac{\partial H}{\partial \lambda} - \dot{x}\right)\delta \lambda \right\} dt$$

$$- \int_0^{t_1^+} \left\{ \left(\frac{\partial H}{\partial u}\right)\delta u + \left(\frac{\partial H}{\partial x} + \dot{\lambda}\right)\delta x + \left(\frac{\partial H}{\partial \lambda} - \dot{x}\right)\delta \lambda \right\} dt.$$

Necessary Conditions,

$$\begin{cases} \lambda(t_1^+) = -\nu \dfrac{\partial \psi}{\partial x(t_1^+)} = -\nu. \\ \lambda(t_1^-) = \nu \dfrac{\partial \psi}{\partial x(t_1^-)} = -\nu. \\ H(t_1^-) = H(t_1^+) \quad : \quad H \text{ is continuous}. \\ \dfrac{\partial H}{\partial u} = 0. \\ \dot{\lambda} = -\left(\dfrac{\partial H}{\partial x}\right)^T \\ \dot{x} = \dfrac{\partial H}{\partial \lambda} = u. \\ x(0) = x_0. \\ x(1) = 0. \end{cases}$$

#4. Direct method.

$\dot{x}(t) = -x(t) + u(t)$ $\quad x(0) = 4.$

$$J = x^2(1) + \int_0^1 \frac{1}{2}u^2(t)dt.$$

Define $H = \frac{1}{2}u^2 + \lambda(-x+u)$, $\phi = x^2(1)$

$\Rightarrow J_a = x^2(1) + \int_0^1 (H - \lambda^T f)dt.$

① Guess $u(t)$.

② Forward integration of $\dot{x}(t) = -x(t) + u(t)$ with the initial condition $x(0) = 4$.

③ Compute $\phi(x(1))$ and $\frac{\partial \phi}{\partial x} = 2x(1)$.

$$\Rightarrow \lambda(1) = \left[-\frac{\partial \phi}{\partial x}\right]_{t=1} = 2x(1).$$

④ Backward integration of $\dot{\lambda}(t) = -\frac{\partial H}{\partial x}$ with the terminal condition $\lambda(1)$.

$$\dot{\lambda}(t) = -\frac{\partial H}{\partial x} = -\lambda. \Rightarrow \lambda(t) = \lambda_0 e^{-t}$$

$$\Rightarrow \lambda(1) = \lambda_0 e^{-1}$$

$$\Rightarrow \lambda_0 = \lambda(1)e$$

$$\therefore \lambda(t) = \lambda(1)e^{1-t}.$$

⑤ Compute $\frac{\partial H}{\partial u} = u + \lambda$

If $|u(t) + \lambda(t)| < \varepsilon$ for some positive real number ε, then stop iteration.

If NOT, $\delta u = -\alpha \frac{\partial H}{\partial u}$. where α is given step

⑥ $u(t) = u^{old}(t) + \delta u(t)$.

⑦ Go to ②.

No. 114

시 험 답 안 지

과목명	최적 궤도제어

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성 적 71

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71

#1 $x(0) = [R\ 0.\ 0.\ 0]$
$F_g = Mg_0R^2/r^2(t)$

$x_1 = r$ $\Rightarrow \dot{x}_1 = x_3$
$x_2 = \alpha$ free $\dot{x}_2 = x_4/x_1$
$x_3 = \dot{r}$ $\dot{x}_3 = T\sin\beta - F_g$
$x_4 = r\dot{\alpha}$ $\dot{x}_4 = T\cos\beta$

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1) $\begin{cases} \dot{x}_1 = x_3 \\ \dot{x}_2 = x_4/x_1 \\ \dot{x}_3 = T\sin u - Mg_0R^2/x_1^2 \\ \dot{x}_4 = T\cos u \end{cases}$ 4

5

Period $= 2h = P$
$\omega = 2\pi/2h = 2\pi/P$

2) $J = \int_0^{t_f} 1\, dt$

$H = 1 + \lambda_1 x_3 + \lambda_2(x_4/x_1) + \lambda_3(T\sin u - Mg_0R^2/x_1^2) + \lambda_4 T\cos u$

$\Rightarrow J_a = \cancel{\dots} + \int_0^{t_f}(H - \lambda^T\dot{x})\, dt$

Costate eq.

$\dot{\lambda} = -\frac{\partial H}{\partial x} \Rightarrow \begin{cases} \dot{\lambda}_1(t) = +\lambda_2 x_4/x_1^2 + 2\lambda_3 MgR^2/x_1^3 \\ \dot{\lambda}_2(t) = 0 \\ \dot{\lambda}_3(t) = \lambda_1 \\ \dot{\lambda}_4(t) = \lambda_2/x_1 \end{cases}$ 4

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Control eq.

$\frac{\partial H}{\partial u} = 0 \Rightarrow \lambda_3 T\cos u - \lambda_4 T\sin u = 0$

식 9개
조건 9개

$\Rightarrow \tan u = \lambda_4/\lambda_3$
$\therefore u = \tan^{-1}(\lambda_4/\lambda_3)$ — 1

Boundary Condition
$H(t_f) = 0$ $\lambda_2(t_f) = 0$ — 1
$x_1(t_f) = R + D$
$x_3(t_f) = 0$ 3
$x_4(t_f) = (\pi/1hour)(R+D)$
$x_1(t_0) = R$ 4
$x_2(0) = 0 = x_3(0) = x_4(0)$

3). $\lambda_2(t_f) = 0$ & $\dot{\lambda}_2(t) = 0 \Rightarrow \lambda_2(t) = 0 \Rightarrow \begin{cases} H = 1 + \lambda_1 x_3 + \lambda_3(T\sin u - MgR^2/x_1^2) + \lambda_4 T\cos u \\ \dot{\lambda}_4(t) = 0 \ \& \ \lambda_4(t) = \text{const} \\ \dot{\lambda}_1(t) = 2\lambda_3 MgR^2/x_1^3 \end{cases}$

state 중 하나는 최종시간에서의 값이 정해지지 않았다. 따라서 초기 값을 추정하여 최종 값을 비교하는 Indirect 방법 보다는 direct 방법으로 수치적으로 풀 수 있을 것이다 u를 추정하여 9개 경계 조건을 만족하는 값을 찾아 나가도록 한다 가장 간단한 direct method는 아래와 같다

① Guess $u(t)$

② Forward Integrating $\dot{x} = f(x, u, t)$ with Initial condition $x(0) = x_0$

③ Compute $\lambda(t_f)$ and ψ, $\dot{\psi}$

④ Backward Integrating $\dot{\lambda}(t)$, H_u^{ψ}

⑤ Compute δu

If $\delta u \le \varepsilon$, Stop iteration.

If $\delta u > \varepsilon$, $u(t) = u^{old}(t) + \delta u$.

⑥ Go to ②.

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시 험 답 안 지

과목명	최적궤도제어

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(20 16 년 12 월 19 일 실시)

성 적	

#2. $\dot{x}(t) = ax(t) + u(t)$ $a>0$. $x(0)=x_0$, $x(t_f)=0$. $|u(t)| \le 1$

2-1) ~~$J = \int_{t_0}^{t_f} 1\,dt = t_f$.~~

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$H = 1 + \lambda(ax(t) + u(t)) = 1 + a\lambda x(t) + \lambda u(t)$

If $|u(t)| \le 1$, the constraint for control does not act.

~~$\Rightarrow \frac{\partial H}{\partial u} = 0 \Rightarrow \frac{\partial H}{\partial u} = \lambda = 0. \quad \therefore \lambda(t) = 0$~~

If $\lambda < 0$, $u(t) = 1$ and If $\lambda > 0$, $u(t) = -1$.

~~$\dot{\lambda} = -\frac{\partial H}{\partial x} \Rightarrow \dot{\lambda}(t) = -a\lambda \Rightarrow \lambda(t) = \lambda_0 e^{-at}$.~~

$$\Rightarrow u(t) = \begin{cases} 1 & \text{if } \lambda(t) < 0 \ (\lambda_0 < 0) \\ -1 & \text{if } \lambda(t) > 0 \ (\lambda_0 > 0) \\ \text{undeterminated.} & \text{if } \lambda(t) = 0. \ (\lambda_0 = 0) \end{cases}$$

$\therefore$ Necessary Conditions

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$$u(t) = \begin{cases} 1 & \text{if } \lambda(t) < 0. \ (\lambda_0 < 0) \\ -1 & \text{if } \lambda(t) > 0 \ (\lambda_0 > 0) \\ \text{undefined} & \text{if } \lambda(t) = 0 \ (\lambda_0 = 0) \end{cases}$$

$\lambda(t) = \lambda_0 e^{-at}$

$\dot{x}(t) = ax(t) + u(t)$

$H(t_f) = 0$

$x(0) = x_0$

$x(t_f) = 0$

Let assume $\lambda(t) = 0$ for finite time.

$\Rightarrow H = 1 + 0(ax(t) + u(t)) = 1 \neq 0$. 모순 contradiction

Hence there is no singular interval Let $\lambda(t) \neq 0$

$\Rightarrow H = 1 + \lambda_0 e^{-at}(ax(t) + u(t)) = 0$.

$\Rightarrow -e^{at}/\lambda_0 = ax(t) + u(t) = \dot{x}(t)$

$\Rightarrow x(t) = -\frac{1}{a\lambda_0} e^{at} + C = +\frac{1}{a}\dot{x}(t) + C. \Rightarrow C = \frac{1}{a}(-\dot{x}(t) + ax(t))$.

Since $x(0) = x_0$ and $x(t_f) = 0$,

$x(0) = -\frac{1}{a\lambda_0} + C = x_0$

$x(t_f) = -\frac{1}{a\lambda_0} e^{at_f} + C = 0$.

$\Rightarrow +\frac{1}{a\lambda_0}(e^{at_f} - 1) = x_0 \Rightarrow \lambda_0 = \frac{1}{ax_0}(e^{at_f} - 1)$.

앞면에서 계속

$$\Rightarrow \begin{cases} \Lambda_0 = \frac{1}{ax_0}(e^{at_f}-1) \\ C = x_0 + x_0/(e^{at_f}-1) = x_0\{e^{at_f}/(e^{at_f}-1)\}. \end{cases}$$

If $x_0>0$, $\Lambda_0>0$, thus $u(t)=-1$

Since $u(t) = \dot{x}(t) - ax(t) = -aC.$

$+1 = +ax_0\{e^{at_f}/(e^{at_f}-1)\}$

$\Rightarrow e^{at_f}-1 = ax_0 e^{at_f}$

$\Rightarrow (1-ax_0)e^{at_f} = 1.$

$\Rightarrow e^{at_f} = \frac{1}{1-ax_0}$ $\leftarrow 1 > ax_0.$

$\Rightarrow t_f = -\frac{1}{a}\ln|1-ax_0|$

When $1>ax_0$, $\exists\ t_f.$

If $x_0<0$, $\Lambda_0<0$, thus $u(t)=1$

Since $u(t) = \dot{x}(t) - ax(t) = aC.$

$\Rightarrow e^{at_f}-1 = -ax_0e^{at_f}.$

$\Rightarrow (1+ax_0)e^{at_f} = 1$ $\leftarrow 1>-ax_0$

$\Rightarrow t_f = -\frac{1}{a}\ln|1+ax_0|$

When $|ax_0|<1$, $\exists\ t_f$

2-2) $J = \int_0^{t_f} u^2(t)\,dt$

$H = u^2 + \Lambda(ax+u) = u^2 + a\Lambda x + \Lambda u.$

If $|u|\le 1$, $\frac{\partial H}{\partial u} = 0.$

$\Rightarrow 2u+\Lambda = 0.$ $\Rightarrow u = -\frac{1}{2}\Lambda$ $(|\Lambda|\le 2)$

If $|u|=1$, By Pontryagin's Minimum Principle, H should be minimum

if $\Lambda>2$, $u=-1$ and if $\Lambda<-2$, $u=1.$

Necessary Condition,

$$\Rightarrow u(t) = \begin{cases} 1 & \text{if } \Lambda<-2. \\ -1 & \text{if } \Lambda>2 \\ -\frac{1}{2}\Lambda & \text{if } |\Lambda|\le 2. \end{cases}$$

$\dot{\Lambda} = -\frac{\partial H}{\partial x} \Rightarrow \dot{\Lambda} = -a\Lambda \Rightarrow \Lambda(t) = \Lambda_0 e^{-at}.$

$H(t_f) = 0.$ $x(0) = x_0.$

$\dot{x} = ax+u.$ $x(t_f) = 0.$

No. 3/4

시 험 답 안 지

과목명	최적제어설계

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(2016 년 12 월 14 일 실시)

성 적	

There is no singular Interval.

2-3) 2-1)은 시간을 최소화하는 문제로 u가 제한 범위의 끝 값만 나오는 bang-bang 문제이다
또한 H가 u에 대해 선형 함수이기 때문에 얼핏보면 제한이 작용하지 않는 경우에는 singular 같은데 실제로는 bang-bang 문제라서 제한 범위 끝 값이 아니면 Hamiltonian $H=0$을 유지하지 못한다. 또한 ax_0 값에 따라 t_f를 못 구할 수도 있다

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2-2)는 연료를 최소화 하는 문제로 u가 제한 범위 내에 존재할 수 있다. 즉, 제한 조건이 작용하지 않기도 한다. 또한 singular interval은 존재하지 않으며 $u(t)$의 값은 λ의 범위에 따라 달라진다

두 경우 모두 $\lambda(t)$ 식 (costate 식)은 동일한데 $a>0$이기 때문에 $t\to\infty$이면 $\lambda(t)\to 0$ 이다

$a<0$ 이라면, time optimal control law는 Necessary Condition은 동일하다.

If $x_0>0$, $u(t)=-1$

$\Rightarrow t_f = -\frac{1}{a}\ln|1-ax_0|$

$1-ax_0>1$ 이므로 항상 t_f 값을 얻을 수 있다.

If $x_0<0$, $u(t)=1$

$\Rightarrow t_f = -\frac{1}{a}\ln|1+ax_0|$

$ax_0>0$ 이므로 항상 t_f를 구할 수 있다

$a<0$ 일때 fuel optimal control의 Necessary Condition 도 동일하다.

두 경우 모두 $\lambda(t)$가 발산하는 함수이다

#3. $\dot{x} = x+u$, $x(0)=x_0$, $x(1)=0$

~~$J=\int_0^1 g(x,u)\,dt$~~

At $t_1=0.5$, $x(t_1^+)=2x(t_1^-)+1$

Let $H = g+\lambda(x+u) = g+\lambda x+\lambda u$, $\phi = x(t_1^+)-2x(t_1^-)-1$

Augmented index is, $J_a = \nu\phi(x(t_1^+),x(t_1^-)) + \int_0^{t_1^-}(H-\lambda^T\dot{x})\,dt - \int_1^{t_1^+}(H-\lambda^T\dot{x})\,dt$

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$\Rightarrow \delta J_a = \nu\frac{\partial\phi}{\partial x(t_1^+)}\delta x_1^+ + \nu\frac{\partial\phi}{\partial x(t_1^-)}\delta x_1^- + [\lambda^T]_{t=t_1^+}\delta x_1^+ - [\lambda^T]_{t=t_1^-}\delta x_1^-$

$+\int_0^{t_1^-}\left\{\frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial x}^T+\dot{\lambda}\right)\delta x + \left(\frac{\partial H}{\partial \lambda}-\dot{x}\right)\delta\lambda\right\}dt$

$-\int_0^{t_1^+}\left\{\frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial x}^T+\dot{\lambda}\right)\delta x + \left(\frac{\partial H}{\partial \lambda}-\dot{x}\right)\delta\lambda\right\}dt$

앞면에서 계속 . .

$$= \left\{\mu \frac{\partial \phi}{\partial x(t_1^+)} + \lambda(t_1^+)\right\}\delta x_1^+ + \left\{\mu \frac{\partial \phi}{\partial x(t_1^-)} - \lambda(t_1^-)\right\}\delta x_1^-$$

$$+ \int_0^{t_1^-} \left\{\frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial x} + \dot{\lambda}\right)\delta x + \left(\frac{\partial H}{\partial \lambda} - \dot{x}\right)\delta\lambda\right\} dt$$

$$- \int_{\phi}^{t_1^+} \left\{\frac{\partial H}{\partial u}\delta u + \left(\frac{\partial H}{\partial x} + \dot{\lambda}\right)\delta x + \left(\frac{\partial H}{\partial \lambda} - \dot{x}\right)\delta\lambda\right\} dt$$

Necessary Condition

$$\begin{cases} \lambda(t_1^+) = -\mu \dfrac{\partial \phi}{\partial x(t_1^+)} = -\mu \\ \lambda(t_1^-) = \mu \dfrac{\partial \phi}{\partial x(t_1^-)} = -2\mu \end{cases}$$

$X(0) = X_0 \quad X(1) = 0$

$X(t_1^+) - 2X(t_1^-) - 1 = 0$

$\frac{\partial H}{\partial u} = 0$ for $t \in [0,1] \setminus \{t_1\}$

$\Rightarrow 0 = \frac{\partial g}{\partial u} + \lambda \quad \cdots (A)$

$\dot{\lambda} = -\frac{\partial H}{\partial x} \Rightarrow \dot{\lambda}(t) = -\frac{\partial g}{\partial x} - \lambda$ for $t \in [0,1] \setminus \{t_1\}$

$\dot{x} = X + U$

Since $H = g + \lambda X + \lambda U$

$H(t_1^-) = g(X(t_1^-), U(t_1^-)) + \lambda_1(t_1^-)(X(t_1^-) + U(t_1^-))$

$H(t_1^+) = g(X(t_1^+), U(t_1^+)) + \lambda_1(t_1^+)(X(t_1^+) + U(t_1^+))$

$\quad = g(2X(t_1^-)+1, U(t_1^+)) + \frac{1}{2}\lambda_1(t_1^-)(2X(t_1^-)+1+U(t_1^+))$

$\mu = 0$ 이면 ϕ을 만족하지 않으므로 $\mu \neq 0$ 이다 좌극한, 우극한이 다르기 때문에 따라서 λ는 $t = t_1$ 에서 불연속이다. 또한 $x(t_1^+) = 2X(t_1^-)+1$ 이므로 $t = t_1$에서 $u = \infty$ 이다. 따라서 $g + \lambda U$ 가 상수가 되지 않는 이상 λU가 $t = t_1$에서 무한대로 되기 때문에 $H(t_1)$ 이 $H(t_1^-)$나 $H(t_1^+)$와 같아 질 것이라는 보장이 없다 연속 함수가 아니어도 적분은 가능하므로 g가 $t = t_1$ 에서 연속이라 확신할 수 없다. (A)에서 λ가 (t_1) 에서 불연속이므로 $\frac{\partial g}{\partial u}$ 도 불연속이다...

H는 X, λ, U에 대해서는 연속인 함수이다. 그러나 t에 대해서는 연속성을 보장할 수 없다. $H(t_1^-) = H(t_1^+)$ 를 만족하게 할 조건이 없다.

t_1이 free 였으면 있었을 텐데...

과목명	최적제어

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(20 년 월 일 실시)

성 적	

#4. Let $t = t_f \tau$ for $\tau \in [0,1]$.

$\Rightarrow dt = p^2 d\tau$

$\Rightarrow \frac{d}{dt} = \frac{1}{p^2}\frac{d}{d\tau}$

$\Rightarrow J = \int_0^{t_f}(1+u^2)dt = \int_0^1 (1+u^2)p^2 d\tau$

$\dot{x} = -x + x^3 + u \Rightarrow x' = (-x+x^3+u)p^2 \quad (x' = dx/d\tau)$

$\dot{p} = 0$

Define Hamiltonian $H = (1+u^2)p^2 + \lambda_1(-x+x^3+u)p^2 + \lambda_2 0$

$\cancel{\frac{\partial H}{\partial u} = 0 \Rightarrow 2up^2 + \lambda_1 p^2 = 0}$

$\Rightarrow (2u+\lambda_1)p^2 = 0 \quad \therefore u = -\frac{1}{2}\lambda_1 \quad (\because p \neq 0)$

$\dot{\lambda} = -\frac{\partial H}{\partial x} \quad \Rightarrow \dot{\lambda}_1(t) = \lambda_1(1-x^2)p^2$

$\Rightarrow \dot{\lambda}_2(t) = (1+u^2-\lambda_1 x + \lambda_1 x^3 + \lambda_1 u)2p = \dot{z}$

Let $\lambda_1 = \lambda$, $\lambda_2 = z$

$\Rightarrow z = \int_0^1 \frac{\partial H}{\partial p} dt \quad \begin{cases} z(1) = 0 \\ z(0) = 0 \end{cases}$

$$\Rightarrow \begin{cases} \dot{x} = (-x + x^3 - \frac{1}{2}\lambda)p^2 \\ \dot{\lambda} = \lambda(1-x^2)p^2 \\ \dot{z} = 2p(1 + \frac{1}{4}\lambda^2 - \lambda x + \lambda x^3 - \frac{1}{2}\lambda^2) \\ \quad = 2p(1 - \frac{1}{4}\lambda^2 - \lambda x + \lambda x^3) \end{cases}$$

$$\Rightarrow \begin{bmatrix} \Delta x_f \\ \Delta \lambda_f \\ \Delta z_f \end{bmatrix} = \begin{bmatrix} (x^2-1)p^2 & -\frac{1}{2}p^2 & 0 \\ -2x\lambda p^2 & (1-x^2)p^2 & 0 \\ 2p\lambda(x^2-1) & 2p(x^3-x-\frac{1}{2}\lambda) & 0 \end{bmatrix} \begin{bmatrix} \Delta x_0 \overset{0}{} \\ \Delta \lambda_0 \\ \Delta z_0 \overset{0}{} \end{bmatrix} + \begin{bmatrix} 2p(-x+x^3-\frac{1}{2}\lambda) \\ 2p\lambda(1-x^2) \\ 2(1-\frac{1}{4}\lambda^2-\lambda x+\lambda x^3) \end{bmatrix} \Delta p$$

$$\Rightarrow \begin{bmatrix} \Delta x_f \\ \Delta z_f \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}p^2 & 2p(-x+x^3-\frac{1}{2}\lambda) \\ 2p(x^3-x-\frac{1}{2}\lambda) & 2(1-\frac{1}{4}\lambda^2-\lambda x+\lambda x^3) \end{bmatrix} \begin{bmatrix} \Delta \lambda_0 \\ \Delta p \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \Delta \lambda_0 \\ \Delta p \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}p^2 & 2p(-x+x^3-\frac{1}{2}\lambda) \\ 2p(x^3-x-\frac{1}{2}\lambda) & 2(1-\frac{1}{4}\lambda^2-\lambda x+\lambda x^3) \end{bmatrix}^{-1} \begin{bmatrix} \Delta x_f \\ \Delta z_f \end{bmatrix} \quad -(B)$$

$$\Rightarrow \begin{cases} \lambda_0 = \lambda_0^{old} + \Delta\lambda_0 \\ \Delta p = p^{old} + \Delta p \end{cases}$$

① Guess $\lambda_0 = \lambda(t_0)$, p

② Using $\frac{\partial H}{\partial u} = 0$, $\dot{x}, \dot{\lambda}$에 대한 식에서 u 없애고 λ와 x 등으로 바꾸기

$$\begin{cases} \dot{x} = f(x, \lambda, t, p) \\ \dot{\lambda} = g(x, \lambda, t, p) \end{cases}$$

앞면에서 계속

③ $\dot{z}$ 에 대한 부분 x, λ, p 에 대해 정리하기

④ Forward 적분하기

$$\left.\begin{cases} \dot{x} = f(x, \lambda, p, t) \\ \dot{\lambda} = g(x, \lambda, p, t) \\ \dot{z} = h(x, \lambda, p, t) \end{cases}\right] \rightarrow (A)$$

⑤ 최종시간에서 계산한 $x(t_f)$, $z(t_f)$ 와 주어진 $x(t_f)$, $z(t_f)$ 비교하여 Δx_f, Δz_f 얻기

⑥ (A) 선형화 하여 State Transition Matrix 구하기

i) 각 파라미터에 1 넣어서 열 별로 구하거나. (하나씩, 나머지는 0)

ii) 각 파라미터를 하나씩 ε 만큼 변화 시켜 STM 변화 얻기

⑦ (B)와 같이 $\Delta\lambda_0$ 와 Δp 구하고 Tol 보다 작으면 계산 그만, 크면 계산을 이어간다.

$$\lambda_0 = \lambda_0^{old} + \Delta\lambda_0 \quad \& \quad p = p^{old} + \Delta p$$

⑧ ④로 돌아가 반복.